Problem Set 1

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Due: October 9, 2025

Question 1: Education

A school counselor was curious about the average of IQ of the students in her school and took a random sample of 25 students' IQ scores. The following is the data set:

```r
y <- c(105, 69, 86, 100, 82, 111, 104, 110, 87, 108, 87, 90, 94, 113, 112, 98,
    80, 97, 95, 111, 114, 89, 95, 126, 98)
```

1.1: Finding a 90% confidence interval for the average student IQ in school

- **Step 1:** Find the point estimate around which a confidence interval can be constructed. We then take this point estimate and by incorporating a margin of error we estimate a range that we think likely a true population parameter would fall in 90% of the time with repeated sampling. In this case, the point estimate is the average (mean) of IQ scores .

$$\bar{y} = \frac{\sum_i^n y_i}{n} \tag{1}$$

```r
mean_y <- mean(y) #Find Point Estimate
mean_y
```

```
[1] 98.44
```

- **Step 2:** Determine the standard error (i.e how different the population mean is likely to be from the sample given the variability and the sample size).

$$\hat{\sigma}_{\bar{y}} = \frac{S}{\sqrt{n}} = \frac{\sqrt{\frac{\sum(y_i-\bar{y})^2}{n-1}}}{\sqrt{n}} \tag{2}$$

```r
n <- length(y) #Sample size
n
se_y <- sd(y) / sqrt(n) #Standard error, where n found by length of y
se_y
```

```
[1] 25
[1] 2.618575
```

- **Step 3:** Find relevant t-score. Since $n < 30$ it is better to use t-distribution as it accounts for smaller sample sizes. We find the t-score by using the relevant tail probability and the degrees of freedom (df). The tail probability is $\alpha/2$ (as two-tailed interval) where the α significance level is the area under the density function minus the tails i.e. 1 minus the wanted 0.9 confidence level $\alpha/2 = ((1-0.9))/2 = 0.05$ and the $df = n - 1 = 24$

```r
tscore_90 <- qt(0.05, 24, lower.tail = FALSE) #Find t-score
    # where p is ((1-.90)/2) = 0.05, df = n - 1 = 24
tscore_90
```

```
[1] 1.710882
```

- **Step 4:** Finally bringing all the elements together to calculate the lower and upper limits of the interval

$$\hat{y} \pm = (t_{0.05,24} * \hat{\sigma}_{\bar{y}}) \tag{3}$$

```r
lower_90_t <- mean_y - (tscore_90 * se_y) # Lower limit of CI: Mean -
    Margin Error
upper_90_t <- mean_y + (tscore_90 * se_y)  # Upper limit of CI: Mean +
    Margin Error
confint_90 <- c(lower_90_t, upper_90_t) #Final Interval
round(confint_90, 2)
```

```
[1]  93.96 102.92
```

The confidence interval for the average student IQ in school is [93.96,102.92]. If we were to conduct repeated sampling of average IQ scores, we would find that 90% of the time the they would fall between the lower and upper bounds of our interval.

1.2: Hypothesis test with $\alpha = 0.05$ to determine whether the average student IQ in her school is higher than the average IQ score (100) among all the schools in the country

- **Step 1: Assumptions.** In this test we are assuming that the random sample IQ scores data fall into a roughly normal distribution. Since it is a relatively small random sample ($n < 30$) we will use a students t-distribution as it better accounts for the increased uncertainty when estimating the population standard deviation from the sample. The sample mean to hypothesis test is $\bar{y} = 98.44$

```
mean_y
```

```
[1] 98.44
```

- **Step 2: Hypothesis** To determine whether the school average IQ score is higher than the population average we prove by contradiction. We construct a null hypothesis – That the sample average is less than or equal to the population – and attempt to disprove it. If we are able to find sufficient evidence to prove the null is statistically very unlikely, we can reject it and accept the alternative – That the school average IQ is greater than the average IQ scores in the country. Given we are testing a directional statement ('greater than') for our sample mean ($\bar{y}$) and that the population average (μ) is stated to be 100, our null and alternative hypotheses are respectively:

$$H_0 : \bar{y} \leq 100 \quad H_a : \bar{y} > 100 \tag{4}$$

- **Step 3: Test Statistic** We proceed by finding out the relevant test statistic i.e. the distance between the estimate and the parameter value in the null hypothesis. For this we find a t- statistic

$$t^* = \frac{\bar{y} - \mu}{\sigma_{\bar{y}}} = \frac{98.44 - 100}{2.61} = -0.596, df = 24 \tag{5}$$

```
test_stat <- (mean_y - 100)/ se_y # sample mean minus population mean
    divide by standard error
test_stat
```

```
[1] -0.5957439
```

- **Step 4: P-value.** We then determine the $p - value$ i.e a probability value that determines how likely it was that we obtained our test statistic given us assuming the null hypothesis.

$$p = Pr(t \geq \frac{\bar{y} - \mu}{\sigma_{\bar{y}}}) \tag{6}$$

```r
p <- pt(test_stat, 24, lower.tail = FALSE) #prob of t-dis of getting test
    stat, df = n-1 = 24, lower tail false as looking at right side
p
```

```
[1] 0.7215383
```

For this one-tailed test this given our assumption that our average is less than or equal to 100 (null) how likely is it that we are to find the sample mean 98.44?

- **Step 5: Conclusion** The p-value $p = 0.72$ is significantly higher than the $\alpha = 0.05$ level. This means the test statistic is likely to occur under the assumption that the null hypothesis (the mean being less than or equal to 100). Thus, there is not sufficient evidence to reject the H_0, and therefore do not find evidence that the school average IQ score is greater than country average.

Question 2: Political Economy

Researchers are curious about what affects the amount of money communities spend on addressing homelessness. The following variables constitute our data set about social welfare expenditures in the USA.

`State`	*50 states in US*
`Y`	*per capita expenditure on shelters/housing assistance in state*
`X1`	*per capita personal income in state*
`X2`	*Number of residents per 100,000 that are "financially insecure" in state*
`X3`	*Number of people per thousand residing in urban areas in state*
`Region`	*1=Northeast, 2= North Central, 3= South, 4=West*

2.0 Importing and Exploring the dataset `expenditure`.

```
stargazer(expenditure)
```

Table 1:

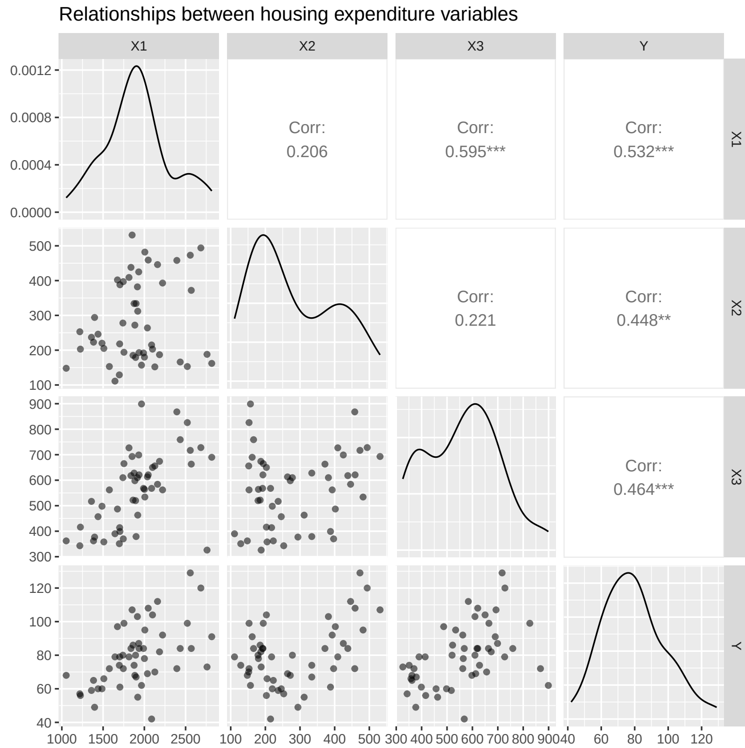
Statistic	N	Mean	St. Dev.	Min	Max
Y	50	79.540	18.507	42	129
X1	50	1,911.900	400.348	1,053	2,817
X2	50	281.780	118.193	111	531
X3	50	561.720	145.037	326	899
Region	50	2.660	1.062	1	4

2.1 Relationships among *Y*, *X1*, *X2*, and *X3*

Plotting the relationships between each pair of variables in the data set, allows exploration of the different associations and patterns of the variation of the explanatory variables with the outcome variable (i.e Y with X1-3) and the explanatory variables with each other.

```
pdf("2.1 Relationships between housing expenditure variables.pdf")
ggpairs(expenditure,
        columns = c(3,4,5,2),
        mapping = aes(alpha = 0.7),
        title = "Relationships between housing expenditure variables")
dev.off()
```

Figure 1: Relationships between US state housing expenditure variables



Density plots: at first glance the density plots of each continuous variable can be seen in the diagonal of the plot matrix. This shows that Y (state expenditure on housing assistance) appears to have a relatively normal distribution. X1 (income pc) is a slightly disjointed bell curve, where X2 (numbers of financially insecure) and X3 (number of urban state residents) both appear bimodal.

Outcome variable Y and predictor variables X's: By looking at the bottom row of the matrix we can see the relevant plots

- **Y and X1:** Plot shows a roughly positive correlation between Y and X1, i.e. an increase in Y and is associated with an increase in X1. At the lower levels of X1 there appears to be a more tight association, and at the upper end more variation and a looser pattern. There do also seem to be a couple outliers.

- **Y and X2:** plot appears very general positive correlation between Y and X2. The association looks to be less linear, slightly 'U' shaped and more loosely related. There seems to be a large amount of bunched values at around 150-200 X2 with a decent variation in associated Y, further illustrating a non-linear relationship potentially.

- **Y and X3:** plot shows positive correlation between Y and X3. The association is loose but relatively more linear, with more similar level's of variation along the X3 axis. There does appear to be some bunching of observations and a couple outliers that fall away from the general trend.

- Overall high levels of state expenditure for housing assistance per capita is loosely associated with higher state personal income per capita, rates of residents with financial insecurity and more urban-dense state demographics.

Relationship between predictor variables: shown in the 3 plots in the middle rows of the matrix

- **X1 and X2:** There does not appear to be a clear correlation between the X1 and X2. There appears to be a gap and between low and high values of X2 that does not variate with the X1. The split in the data indicates the if there is a correlation it is be non-linear and needs further investigation.

- **X1 and X3:** it appears a positive correlation between X1 and X3 meaning that states with higher levels of personal income per capita are associated with higher levels of urban residency. The trend is fairly consistent except for two notable outliers, one in the middle the X1 with substantially higher X3 than the trend suggests and one at the near top of X1 with one of the lowest values of X3 despite the trend predicting it out be near the top.

- **X2 and X3:** there does not appear to be a correlation between X2 and X3. The spread of points very loose and does not show a linear relationship between the two variables. There appears to be a large amount of states with the same level of X2 (at around 150) that have a wide range of corresponding X3, giving evidence that there is not a clear variance relationship with them.

- This correlation is important to consider in their relationship with state housing assistance expenditure and it may mean there is a confounding variable impacting both of the predictor variables and their association with the outcome.

2.2 Relationship between *Y* and *Region*.

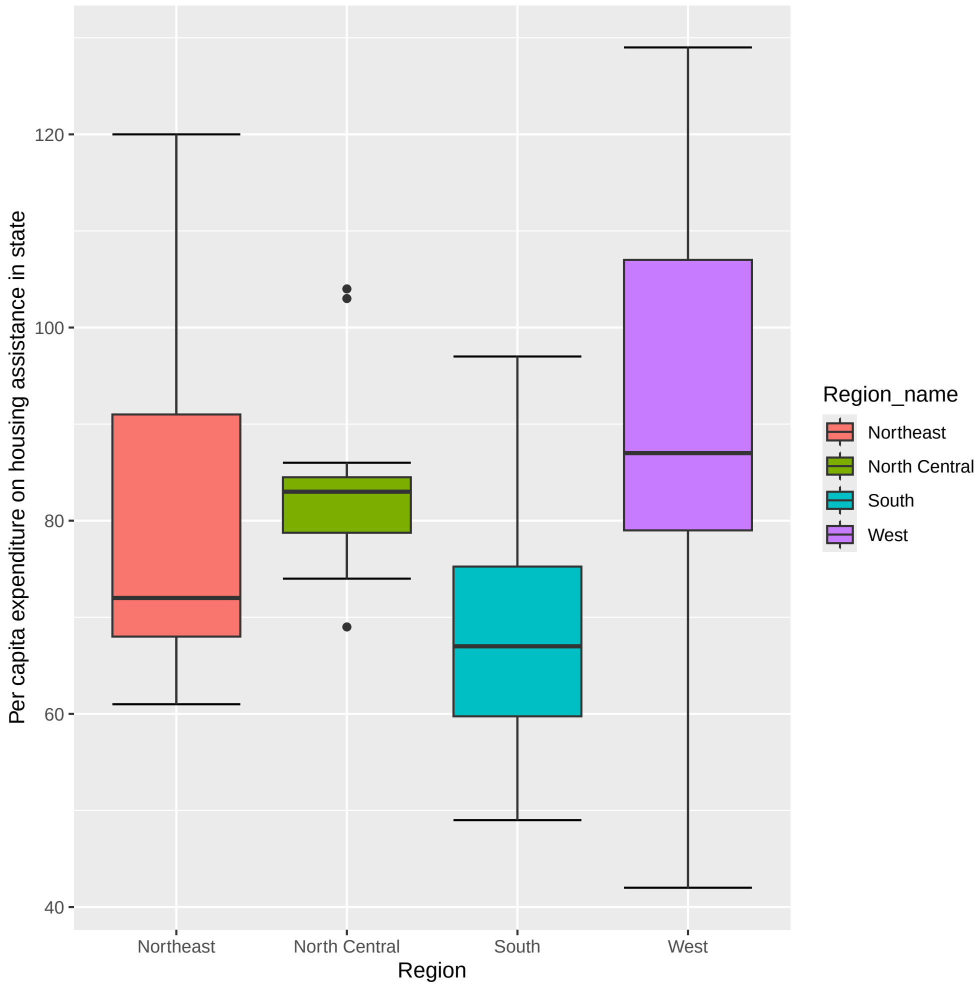
A boxplot is a good way to explore the relationship between state expenditure on housing assistance per capita by region as it allows easy side-by-side comparisons of the range, interquartile range and mean of each regional category.

```r
expenditure$Region_name <- factor(expenditure$Region,
                                  labels = c("Northeast", "North Central", "South", "West"))
# Create new factor variable into Categories from encoding in data frame

pdf("2.2 State Housing Expenditure by Region Boxplot.pdf")
ggplot(expenditure, aes(x = Region_name, y = Y, fill = Region_name)) + #fill colour by region category
```

```
  stat_boxplot(geom ='errorbar') + #to add whiskers
  geom_boxplot() +
  labs(x = "Region", y = "Per capita expenditure on housing assistance in
    state")
dev.off()
```

Figure 2: US region and in state per capita expenditure on housing assistance



The boxplot shows that states in the West of the US have the largest range of expenditure on housing including states with the highest and the lowest expenditure per capita respectively. Southern states seem to have on average the lowest expenditure (lowest IQR and mean) whilst West despite the variability on average has the highest expenditure. North central looks to have the second highest average expenditure and with much less variation within values (with a small box IQR) but does have a few outliers. Northeast region is third highest. To confirm, the average expenditure of each region is found and displayed below. This confirms the West has the greatest on average expenditure on housing assistance per capita with a average value of 88.31.

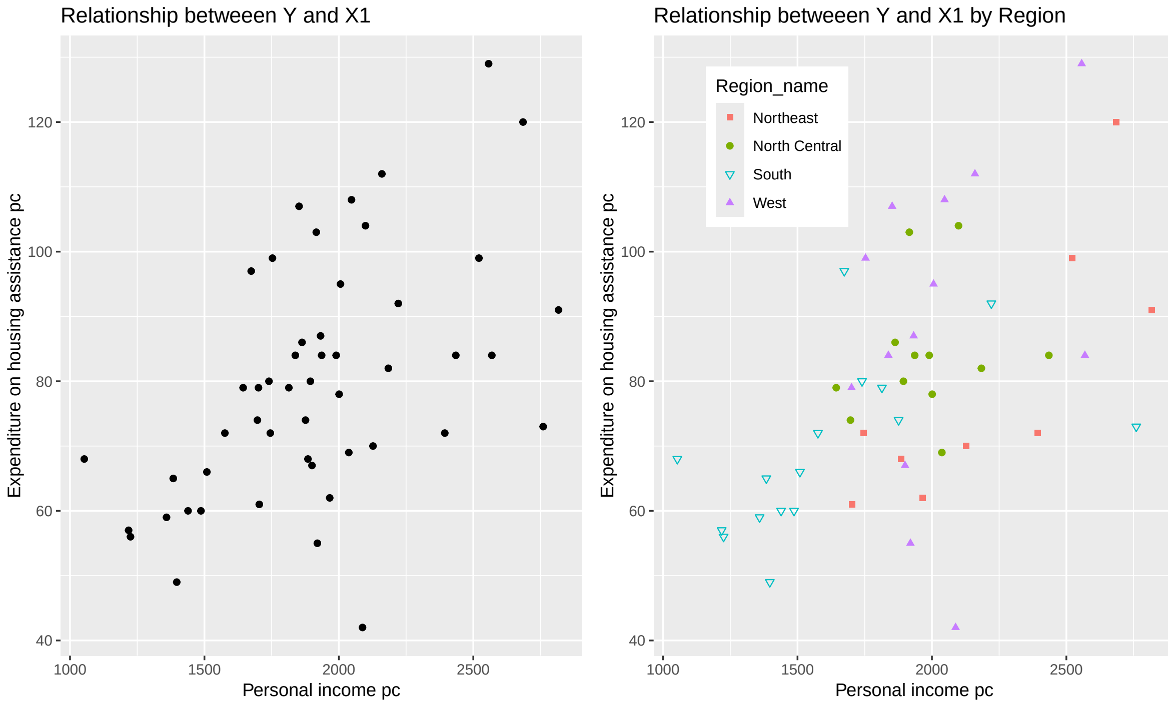
```r
mean_by_region <- aggregate(expenditure$Y,
                            by = list(Region = expenditure$Region_name),
                            FUN = mean,
                            ) #find summary stats for subset Y and region with function mean
mean_by_region #Print region averages
```

```
         Region        x
1     Northeast 79.44444
2 North Central 83.91667
3         South 69.18750
4          West 88.30769
```

2.3 Relationship between *Y* and *X1* then including variable Region

```r
plot1 <- ggplot(expenditure, aes(x = X1, y = Y)) + #using only X1
  geom_point() +
  labs(x = "Personal income pc",
       y = "Expenditure on housing assistance pc",
       title = "Relationship betweeen Y and X1")
plot2 <- ggplot(expenditure, aes(x = X1, y =Y)) +
  geom_point(mapping = aes(color = Region_name, shape = Region_name)) + #distinguish region by mapping points differently
  scale_shape_manual(values=c(15, 19, 6, 17)) + #specify different looking shapes
  labs(x = "Personal income pc",
       y = "Expenditure on housing assistance pc",
       title = "Relationship betweeen Y and X1 by Region") +
  theme(legend.position = c(.1,.95), #legend location
        legend.justification = c("left", "top"),
        legend.box.just = "left",
        legend.margin = margin(6, 6, 6, 6))
plotcom <-  grid.arrange(plot1, plot2, ncol = 2) #combine plots to one grid
ggsave(filename="2.3 Combined plot.pdf", plot= plotcom , width=10, height=6, units="in") #combining together
dev.off()
```

Figure 3: Relationship between housing assistance expenditure and personal income per capita in US states



- **Relationship between Y and X1:** the left figure shows the positive correlation between Y and X1 as briefly explored in 2.1 i.e greater values of X1 are seen to have greater values of Y. The relationship seems linear but relatively loose with many observations spread out at similar levels of X1. At higher levels of X1 there seems to be more variation and spread away from a tight trend. A few outliers are can also be highlighted in particular the lowest Y observation having a much larger X1 per capita personal income in state than would be predicted by a linear trend.

- **Including region:** the right plot in the figure shows the same relationship but incorporates Region by showing each point as a regional specific shape/colour combination (described in the legend). This allows us to see that there is some level of grouping/trends within the Y, X1 relationship by region. There is a collection at the low values of housing expenditure and personal income in state of states in the South of the USA. There is a grouping of North Central states around the middle of the graph relatively close together. It is clear in this case to see the variation of states in the West with the highest and lowest value in housing expenditure but also the greatest number of points that exceed the general trending line on exceeding expenditure on housing than the national trend given the per capita personal income in state.